

ChemDoE Mathematical Foundations

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1 Introduction

When performing a *Design of Experiments* (DoE), the goal is to model and explore the relationship between a set of independent variables (factors) and one or more responses y . The resulting mathematical representation is referred to as a regression model. In practice, the true functional relationship between the response and the factors is typically unknown; therefore, within the experimental region it is common to approximate it using low-order polynomial models. For example, a scientist may be interested in linking reaction yield to temperature and pressure, and then use this empirical model for prediction or optimization.

A general formulation of the multiple linear regression (MLR) model expresses the response as a linear combination of k model terms

$$y = \beta_0 + \sum_{j=1}^k \beta_j x_j + \varepsilon, \quad (1)$$

where x_j denotes the j -th model term (e.g. a main effect, an interaction term, or a higher-order transformation of the original factors), β_0 is the intercept, $\{\beta_1, \dots, \beta_k\}$ are unknown coefficients, and ε is a random error term capturing experimental variability (e.g. measurement noise and other unmodeled sources), typically assumed to satisfy $\mathbb{E}[\varepsilon] = \mathbf{0}$ and $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$.

Equivalently, for n experimental runs, the model can be written in matrix form as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (2)$$

where $\mathbf{y} \in \mathbb{R}^n$ is the response vector, $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the model matrix (including the intercept column), $\boldsymbol{\beta} \in \mathbb{R}^p$ is the parameter vector, and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector.

2 Matrix notation in DoE

In *Design of Experiments* (DoE), matrix notation provides a compact way to describe both the experimental plan and the regression model used for analysis.

Assume that n experimental runs are carried out by varying m controllable factors (for instance temperature, pressure, or pH). The factor settings are col-

lected in the *design matrix*

$$\mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \cdots & d_{nm} \end{bmatrix} \in \mathbb{R}^{n \times m},$$

where each row corresponds to one run and each column to one factor. In practice, $\mathbf{D}$ may contain either factor values in natural units or coded levels (e.g. mapping each factor to a standardized range such as $[-1, 1]$).

Each run produces one or more measured responses (e.g. yield, impurity, intensity). A single response can be stored in a vector $\mathbf{y} \in \mathbb{R}^n$, while l responses can be collected in the response matrix

$$\mathbf{Y} = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1l} \\ y_{21} & y_{22} & \cdots & y_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nl} \end{bmatrix} \in \mathbb{R}^{n \times l}.$$

The i -th row $\mathbf{y}_{i,:} \in \mathbb{R}^{1 \times l}$ contains all responses measured in run i , while the j -th column $\mathbf{y}_{:,j} \in \mathbb{R}^n$ contains the j -th response across the whole design.

For regression, the raw factor settings in $\mathbf{D}$ are typically not used directly. Instead, they are expanded into a set of *model terms* according to the chosen model structure (intercept, main effects, interactions, quadratic terms, *etc.*). Collecting these terms for all runs yields the *model matrix* $\mathbf{X}$. Let k denote the number of non-intercept model terms, so that the total number of columns is $p = k + 1$. Then

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \in \mathbb{R}^{n \times p}, \quad p = k + 1.$$

The model matrix can be constructed by simple rules: the first column is the intercept (a column of ones), interaction terms are obtained by multiplying the corresponding columns, and quadratic terms are obtained by squaring the corresponding columns.

For the j -th response $\mathbf{y}_{:,j}$, the coefficient vector is

$$\boldsymbol{\beta}_j = \begin{bmatrix} \beta_{0j} \\ \beta_{1j} \\ \vdots \\ \beta_{kj} \end{bmatrix} \in \mathbb{R}^p,$$

and stacking all coefficient vectors gives the coefficient matrix

$$\mathbf{B} = \begin{bmatrix} \beta_{01} & \beta_{02} & \cdots & \beta_{0l} \\ \beta_{11} & \beta_{12} & \cdots & \beta_{1l} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{k1} & \beta_{k2} & \cdots & \beta_{kl} \end{bmatrix} \in \mathbb{R}^{p \times l}.$$

The error terms can be collected in an error matrix $\mathbf{E} \in \mathbb{R}^{n \times l}$, whose j -th column is $\boldsymbol{\varepsilon}_j \in \mathbb{R}^n$. The multivariate regression model is therefore written as

$$\mathbf{Y} = \mathbf{XB} + \mathbf{E}. \quad (3)$$

After estimating $\mathbf{B}$, the fitted responses are $\hat{\mathbf{Y}} = \mathbf{XB}$, and the residual matrix is $\hat{\mathbf{E}} = \mathbf{Y} - \hat{\mathbf{Y}}$.

In many DoE applications, responses are still fitted and interpreted separately (column-wise), but the stacked notation in Eq. (3) is useful to keep the discussion compact and to highlight the common structure shared by all responses.

3 Linear Regression Models

From now on, we will develop the theory for a single response. The extension to multiple responses is straightforward and follows the same reasoning. In typical MLR, each response is modeled and fitted independently, a separate regression model is estimated for each response variable.

To illustrate the main ideas, consider a simple case where the goal is to model the reaction yield as a function of temperature and pressure. A first-order (main-effects only) model can be written as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon, \quad (4)$$

where y is the yield, x_1 the temperature, x_2 the pressure, and ε is a random error term. This is a *linear* regression model because it is linear in the unknown parameters $\beta_0, \beta_1, \beta_2$. In general, the linear model can then be written as

$$y = \beta_0 + \sum_{j=1}^p \beta_j x_j + \varepsilon, \quad (5)$$

where $\{\beta_0, \beta_1, \dots, \beta_k\}$ are the *regression coefficients*.

More complex models are obtained by using more model terms. For example, a two-factor interaction between x_1 and x_2 can be introduced

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon, \quad (6)$$

where β_{12} quantifies the interaction between x_1 and x_2 .

A common choice in DoE is the second-order (quadratic) model, which also captures curvature via squared terms:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \varepsilon. \quad (7)$$

Extending this idea to a set of p model terms yields the general second-order linear model

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \sum_{j>i}^k \beta_{ij} x_i x_j + \sum_{i=1}^k \beta_{ii} x_i^2 + \varepsilon, \quad k = p - 1 \quad (8)$$

The key point is that the model is *linear in the parameters*, even if the model terms themselves are nonlinear functions of the original factors. In some applications, higher-order terms (e.g., three-factor interactions $\beta_{ijk} x_i x_j x_k$ or cubic terms $\beta_{iii} x_i^3$) may also be considered, although they typically require larger designs and are therefore used less frequently in practice. The main goal is therefore to estimate the parameter vector $\boldsymbol{\beta} \in \mathbb{R}^p$ from the experimental data.

4 Ordinary Least Squares (OLS) Estimation

A standard approach to estimate the parameters of a multiple linear regression model is the *least squares* (LS) method. Assume the model written in matrix notation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{y} \in \mathbb{R}^n, \mathbf{X} \in \mathbb{R}^{n \times p}, \boldsymbol{\beta} \in \mathbb{R}^p, \boldsymbol{\varepsilon} \in \mathbb{R}^n. \quad (9)$$

As specified in the previous section, $p = k + 1$, where k is the number of non-intercept model terms and the additional term accounts for the intercept.

The LS framework relies on the following common assumptions:

- $\mathbf{X}$ has full column rank, i.e. $\text{rank}(\mathbf{X}) = p$ (in particular, $n \geq p$);
- the model structure (i.e. the set of model terms in $\mathbf{X}$) is correctly specified within the experimental region;
- the error term satisfies $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ (uncorrelated errors with constant variance).

Written observation-wise, the model reads

$$y_i = \beta_0 + \sum_{j=1}^k \beta_j x_{ij} + \varepsilon_i, \quad i = 1, \dots, n, \quad (10)$$

where x_{ij} is the value of the j -th model term at run i . Given a generic parameter vector $\boldsymbol{\beta}$, the fitted values are

$$\hat{\mathbf{y}}(\boldsymbol{\beta}) = \mathbf{X}\boldsymbol{\beta}, \quad \hat{y}_i(\boldsymbol{\beta}) = \mathbf{x}_i^\top \boldsymbol{\beta},$$

where $\mathbf{x}_i^\top$ denotes the i -th row of $\mathbf{X}$. The corresponding residual vector is

$$\mathbf{e}(\boldsymbol{\beta}) = \mathbf{y} - \hat{\mathbf{y}}(\boldsymbol{\beta}) = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}.$$

The LS criterion minimizes the sum of squared residuals

$$L(\boldsymbol{\beta}) = \mathbf{e}(\boldsymbol{\beta})^\top \mathbf{e}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (11)$$

Expanding the product gives

$$L(\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}. \quad (12)$$

4.1 Normal equations

The least squares estimate of $\boldsymbol{\beta}$ is defined as

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2. \quad (13)$$

Differentiating $L(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ yields

$$\nabla_{\boldsymbol{\beta}} L(\boldsymbol{\beta}) = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

Setting the gradient to zero gives the *normal equations*

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}. \quad (14)$$

If $\mathbf{X}$ has full column rank (equivalently, if $\mathbf{X}^\top \mathbf{X}$ is invertible), the solution is unique and is given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (15)$$

The matrix $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{p \times p}$ is the *Gram matrix* of the columns of $\mathbf{X}$. Denoting by $\mathbf{x}_j \in \mathbb{R}^n$ the j -th column of $\mathbf{X}$, its elements are

$$(\mathbf{X}^\top \mathbf{X})_{ij} = \mathbf{x}_i^\top \mathbf{x}_j, \quad i, j = 0, \dots, k. \quad (16)$$

Diagonal entries $\mathbf{x}_j^\top \mathbf{x}_j$ are sums of squares of each model term, whereas off-diagonal entries $\mathbf{x}_i^\top \mathbf{x}_j$ are cross-products that quantify the *non-orthogonality* between model terms.

Once $\hat{\boldsymbol{\beta}}$ is available, the fitted responses are

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}}, \quad (17)$$

and the residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}} \quad (18)$$

which is the difference between the actual observation y_i and the fitted value $\hat{y}_i$. By construction, LS residuals are orthogonal to the column space of $\mathbf{X}$, i.e. $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.

4.2 Error vs. residuals

In the regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

the vector $\boldsymbol{\varepsilon}$ represents the *errors*, i.e. random deviations from the true mean response due to experimental variability (e.g. measurement noise and run-to-run fluctuations). Since $\boldsymbol{\beta}$ is unknown, the errors $\boldsymbol{\varepsilon}$ are not directly observable.

After fitting the model, we obtain the estimated coefficients $\hat{\boldsymbol{\beta}}$ and the fitted responses

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

The *residuals* are then defined as the observable discrepancies between measured and fitted values:

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Residuals are therefore *estimates* of the (unobserved) errors. Substituting the model into the definition of $\mathbf{e}$ yields

$$\mathbf{e} = (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) - \mathbf{X}\hat{\boldsymbol{\beta}} = \boldsymbol{\varepsilon} - \mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}),$$

showing that residuals differ from errors by a term that depends on the coefficient estimation error.

From a modeling perspective, it is also useful to distinguish between (i) random experimental variation and (ii) model inadequacy. Let $f(\mathbf{d}_i)$ denote the true (unknown) mean response at factor setting $\mathbf{d}_i$, and let η_i be random experimental noise with $\mathbb{E}[\eta_i] = 0$. If $\mathbf{x}_i$ denotes the regressor vector constructed from $\mathbf{d}_i$ and the chosen model terms, then

$$y_i = f(\mathbf{d}_i) + \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \delta_i + \eta_i,$$

where $\delta_i = f(\mathbf{d}_i) - \mathbf{x}_i^\top \boldsymbol{\beta}$ represents the model discrepancy (lack-of-fit). In the classical linear model, it is typically assumed that $\delta_i = 0$ within the experimental region; under this assumption, the error term reflects only random experimental variability and standard OLS inference applies.

4.3 Residual sum of squares

A key quantity for assessing the fit is the *residual sum of squares* (SS_E), defined as

$$\text{SS}_E = \mathbf{e}^\top \mathbf{e} = \sum_{i=1}^n e_i^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|_2^2. \quad (19)$$

Since $\hat{\boldsymbol{\beta}}$ minimizes the loss function, SS_E is the minimum value of $L(\boldsymbol{\beta})$.

Under the classical assumptions $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$, an unbiased estimator of the error variance (σ^2) is obtained by scaling the SS_E by its degrees of freedom:

$$\hat{\sigma}^2 = \frac{\text{SS}_E}{n - p}, \quad (20)$$

where $n - p$ is the residual degrees of freedom (number of observations n minus number of estimated coefficients p , including the intercept). The corresponding quantity

$$s = \sqrt{\hat{\sigma}^2} \quad (21)$$

is often referred to as the *residual standard error* (or root mean squared error) of the fit.

4.4 Variance and covariance of the coefficient estimates β

Under the classical assumptions $\mathbb{E}[\varepsilon] = \mathbf{0}$ and $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$, the least squares estimator

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

is unbiased, i.e. $\mathbb{E}[\hat{\beta}] = \beta$, and its covariance matrix is

$$\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (22)$$

In practice, σ^2 is unknown and is replaced by its unbiased estimate

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p}, \quad (23)$$

yielding the estimated covariance matrix

$$\widehat{\text{Cov}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (24)$$

The variance of the individual coefficient estimate $\hat{\beta}_j$ is given by the j -th diagonal element of Eq. (22):

$$\text{Var}(\hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}, \quad j = 0, \dots, p - 1, \quad (25)$$

and the covariance between two coefficient estimates $\hat{\beta}_i$ and $\hat{\beta}_j$ is given by the corresponding off-diagonal element:

$$\text{Cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{ij}. \quad (26)$$

Large off-diagonal entries indicate strong correlation between coefficient estimates, which typically arises when model terms are not orthogonal. In DoE, designs are often constructed to reduce such correlations, improving the precision and interpretability of the estimated effects.

The square root of the diagonal elements of the estimated covariance matrix gives the standard errors of the regression coefficients:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}} = \sqrt{\frac{\text{SSE}}{n - p} [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}. \quad (27)$$

Orthogonality. Orthogonal designs are particularly attractive because they simplify interpretation and reduce correlation among coefficient estimates. If the columns of $\mathbf{X}$ are mutually orthogonal, i.e. $\mathbf{x}_i^\top \mathbf{x}_j = 0$ for $i \neq j$, then the Gram matrix is diagonal:

$$\mathbf{X}^\top \mathbf{X} = \text{diag}(\mathbf{x}_0^\top \mathbf{x}_0, \mathbf{x}_1^\top \mathbf{x}_1, \dots, \mathbf{x}_{p-1}^\top \mathbf{x}_{p-1}),$$

which implies

$$(\mathbf{X}^\top \mathbf{X})_{ij} = (\mathbf{x}_i^\top \mathbf{x}_i) \delta_{ij}, \quad i, j = 0, \dots, p-1. \quad (28)$$

Consequently,

$$[(\mathbf{X}^\top \mathbf{X})^{-1}]_{ij} = \frac{1}{\mathbf{x}_i^\top \mathbf{x}_i} \delta_{ij}, \quad i, j = 0, \dots, p-1, \quad (29)$$

and therefore

$$\text{Cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{ij} = \sigma^2 \frac{1}{\mathbf{x}_i^\top \mathbf{x}_i} \delta_{ij}. \quad (30)$$

Hence, $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j) = 0$ for $i \neq j$, meaning that the coefficient estimates are uncorrelated.

D-optimality. Equation (22) highlights the role of $\mathbf{X}^\top \mathbf{X}$ as an *information matrix*: larger values (in an appropriate matrix sense) correspond to smaller uncertainty in the estimated coefficients. A widely used optimality criterion is *D-optimality*, which selects the experimental design (i.e. the rows of $\mathbf{X}$) to maximize the determinant of the information matrix,

$$\max \det(\mathbf{X}^\top \mathbf{X}). \quad (31)$$

Equivalently, up to the multiplicative constant σ^{2p} , this criterion minimizes the determinant of the covariance matrix of the coefficient estimates:

$$\det(\text{Cov}(\hat{\beta})) = \sigma^{2p} \det((\mathbf{X}^\top \mathbf{X})^{-1}) = \frac{\sigma^{2p}}{\det(\mathbf{X}^\top \mathbf{X})}.$$

Thus, D-optimal designs minimize the *generalized variance* of $\hat{\beta}$ and, in aggregate, produce the most precise parameter estimates among competing designs of the same size.

5 Hypothesis testing in MLR

In this section we review the main hypothesis testing procedures in multiple linear regression. These procedures typically assume that the error terms are independent and normally distributed with mean zero and constant variance:

$$\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n.$$

Equivalently, in vector form,

$$\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

5.1 Test for Significance of Regression

A fundamental question is whether the regression model explains a statistically significant portion of the variability in the response. The *test for significance of regression* compares the fitted model against the intercept-only model, to determine if a linear relationship exists between the response variable y and a subset of model terms $\{x_1, x_2, \dots, x_k\}$.

Hypotheses. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ include the intercept (so $p = k + 1$). The global null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0 \quad (32)$$

while the alternative hypothesis is

$$H_1 : \text{at least one } \beta_j \neq 0 \ (j \geq 1), \quad (33)$$

i.e., all slope coefficients are zero and the mean response is constant (only the intercept is needed). If we remember the form of the predicted response vector $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ and the sample mean as $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$, we can define the following quantities (ANOVA decomposition)

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2 \quad (\text{total sum of squares}), \quad (34)$$

$$SS_E = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{residual sum of squares}), \quad (35)$$

$$SS_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \quad (\text{regression sum of squares}). \quad (36)$$

The variability decomposes as

$$SS_T = SS_R + SS_E. \quad (37)$$

The total sum of squares SS_T can be partitioned into a sum of squares due to the model (or to regression) - SS_R and a sum of squares due to residual - SS_E .

Mean squares and F statistic. The regression and error mean squares are defined as

$$MS_R = \frac{SS_R}{p-1}, \quad MS_E = \frac{SS_E}{n-p}. \quad (38)$$

The test statistic for the global significance of regression is

$$F = \frac{MS_R}{MS_E} = \frac{SS_R/(p-1)}{SS_E/(n-p)}. \quad (39)$$

Under H_0 and the normality assumptions, the statistic in Eq. (39) follows an F distribution with $(p-1, n-p)$ degrees of freedom:

$$F \sim F_{p-1, n-p}.$$

The null hypothesis is rejected at significance level α if $F > F_{1-\alpha; p-1, n-p}$ (equivalently, if the associated p -value is smaller than α). Rejecting H_0 indicates that the model provides a statistically significant improvement over the intercept-only model.

Connection to R^2 . Using Eq. (37), the coefficient of determination is

$$R^2 = \frac{SS_R}{SS_T} = 1 - \frac{SS_E}{SS_T}. \quad (40)$$

A large F statistic corresponds to a large fraction of explained variability (large R^2) relative to the residual variability. The problem with R^2 is that an high value doesn't imply a good regression model, since adding a variable to the model will always increase R^2 . This can be partially avoided by using the adjusted R^2 statistic

$$R_{adj}^2 = 1 - \frac{SS_E/(n-p)}{SS_T/(n-1)} = 1 - \left(\frac{n-1}{n-p}\right)(1 - R^2) \quad (41)$$

DoE interpretation. In a DoE context, this global test is often used as a first check that the chosen set of model terms (e.g., main effects, interactions, quadratic terms) captures systematic changes in the response beyond experimental noise.

5.2 Variance Inflation Factor (VIF).

When multiple regressors are correlated, the estimated regression coefficients may become unstable and their standard errors can increase, even when the overall model fit remains satisfactory. A standard diagnostic for this phenomenon is the *Variance Inflation Factor* (VIF). For the j -th regressor, the VIF is defined as

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (42)$$

where R_j^2 is the coefficient of determination obtained by regressing x_j on all the remaining regressors in the model. The quantity VIF_j measures how much the variance of $\hat{\beta}_j$ is inflated due to collinearity relative to the ideal case of orthogonal predictors; in particular,

$$\text{Var}(\hat{\beta}_j) = \sigma^2 \frac{1}{S_{x_j x_j}} \text{VIF}_j, \quad (43)$$

with $S_{x_j x_j} = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$ and σ^2 the error variance. Values of VIF_j close to 1 indicate negligible collinearity, whereas large values suggest that $\hat{\beta}_j$ may be poorly determined and sensitive to small perturbations of the data. In the context of DoE, low VIFs are typically associated with well-conditioned model matrices and hence with designs that provide more reliable coefficient estimates.

5.3 Test for Model Terms Significance

After assessing the global significance of the regression, it is often of interest to determine which individual model terms contribute significantly to explaining the response. This is typically done by testing each coefficient separately while keeping the remaining terms in the model fixed.

Hypotheses. For a given model term indexed by $j \in \{1, \dots, p-1\}$ (excluding the intercept), the hypothesis test is

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0. \quad (44)$$

Test statistic. Under the assumptions $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ and $\text{rank}(\mathbf{X}) = p$, the estimated covariance matrix of $\hat{\beta}$ is

$$\widehat{\text{Cov}}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \hat{\sigma}^2 = \frac{\text{SSE}}{n-p}.$$

Hence, the standard error of $\hat{\beta}_j$ is

$$\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}.$$

The corresponding t statistic is

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}. \quad (45)$$

Decision rule. Under H_0 , the statistic in Eq. (45) follows a Student t distribution with $n-p$ degrees of freedom:

$$t_j \sim t_{n-p}.$$

The null hypothesis is rejected at significance level α if

$$|t_j| > t_{1-\alpha/2; n-p},$$

or equivalently if the associated p -value is smaller than α . Rejecting H_0 suggests that the corresponding model term has a statistically significant effect on the response, conditional on the other terms included in the model.

DoE interpretation. In DoE, these tests are commonly used to screen out negligible effects and to identify the most influential model terms. However, when the design is not orthogonal (or when higher-order terms are included), coefficient estimates can be correlated; in this case, term significance should be interpreted with care and in conjunction with the global F -test and diagnostic checks.

5.4 Test for Lack of Fit

When replicated observations are available at one or more design points, it is possible to assess whether the assumed regression model adequately describes the true relationship between the response and the model terms. This assessment is carried out through the **lack-of-fit test**, which separates the residual variability into components due to pure experimental error and model inadequacy. Suppose that observations are taken at m distinct regressor settings $\mathbf{x}_i$, with n_i replicated measurements at the i -th setting, for $i = 1, 2, \dots, m$. Let y_{ij} denote the j -th observation at $\mathbf{x}_i$, with $j = 1, 2, \dots, n_i$, and let

$$n = \sum_{i=1}^m n_i$$

be the total number of observations. Denote by $\bar{y}_i$ the sample mean of the responses at $\mathbf{x}_i$, and by $\hat{y}_i$ the fitted value provided by the regression model at the same point. The (i, j) -th residual can be decomposed as

$$y_{ij} - \hat{y}_i = (y_{ij} - \bar{y}_i) + (\bar{y}_i - \hat{y}_i). \quad (46)$$

Squaring both sides and summing over all observations yields the partition of the residual sum of squares:

$$SS_E = SS_{PE} + SS_{LOF}, \quad (47)$$

where SS_E is the usual residual sum of squares, SS_{PE} is the sum of squares due to pure error, and SS_{LOF} is the sum of squares due to lack of fit.

The pure error sum of squares is defined as

$$SS_{PE} = \sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2, \quad (48)$$

and represents a *model-independent* estimate of experimental variability, as it depends only on the dispersion of replicated observations at identical regressor settings. The associated degrees of freedom are

$$\nu_{PE} = \sum_{i=1}^m (n_i - 1) = n - m. \quad (49)$$

The lack-of-fit sum of squares is given by

$$SS_{LOF} = \sum_{i=1}^m n_i (\bar{y}_i - \hat{y}_i)^2, \quad (50)$$

which measures the discrepancy between the fitted model and the average response at each design point. This component reflects systematic deviations

that cannot be attributed to random experimental error alone. The degrees of freedom associated with lack of fit are

$$\nu_{LOF} = m - p, \quad (51)$$

where p is the number of estimated regression parameters, including the intercept. The hypotheses for the lack-of-fit test are

H_0 : the regression model fits the data adequately,

H_1 : the regression model does not fit the data adequately.

The test statistic is constructed as

$$F_0 = \frac{SS_{LOF}/(m - p)}{SS_{PE}/(n - m)} = \frac{MS_{LOF}}{MS_{PE}}, \quad (52)$$

which, under the null hypothesis, follows an F distribution with $(m - p, n - m)$ degrees of freedom. If F_0 exceeds the critical value $F_{\alpha, m-p, n-m}$, there is evidence of lack of fit and the assumed regression model should be reconsidered. Conversely, failure to reject H_0 indicates that any deviations between the fitted model and the observed data are consistent with pure experimental error.

6 Confidence Intervals in MLR

In addition to point estimation and hypothesis testing, an important aspect of multiple linear regression (MLR) analysis is the quantification of uncertainty associated with estimated parameters and predicted responses. This uncertainty is typically expressed through **confidence intervals**, which provide a range of plausible values for unknown quantities based on the observed data and a specified confidence level.

In the context of MLR, confidence intervals can be constructed for individual regression coefficients, for the mean response at a given point in the experimental region, and for a new future observation. Each of these intervals serves a different inferential purpose and has a distinct interpretation.

6.1 Confidence Intervals on Individual Coefficients

Consider the MLR model written in matrix form as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (53)$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. Under these assumptions, the least squares estimator $\hat{\boldsymbol{\beta}}$ is normally distributed:

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}). \quad (54)$$

Since σ^2 is generally unknown, it is replaced by its unbiased estimator

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p}, \quad (55)$$

where n is the number of observations and p is the number of model parameters.

A $(1 - \alpha)100\%$ confidence interval for the j -th regression coefficient β_j is given by

$$\hat{\beta}_j \pm t_{\alpha/2, n-p} \hat{\sigma} \sqrt{(\mathbf{X}^\top \mathbf{X})_{jj}^{-1}}, \quad (56)$$

where $t_{\alpha/2, n-p}$ denotes the upper $\alpha/2$ quantile of the Student's t distribution with $n - p$ degrees of freedom.

This interval provides a range of plausible values for β_j and is commonly used to assess both the magnitude and the practical relevance of individual model terms.

6.2 Confidence Intervals on the Mean Response

Let $\mathbf{x}_0$ be a vector of regressor values corresponding to a specific point in the experimental region, including the intercept term. The mean response at $\mathbf{x}_0$ is defined as

$$\mu(\mathbf{x}_0) = \mathbb{E}[y \mid \mathbf{x}_0] = \mathbf{x}_0^\top \boldsymbol{\beta}. \quad (57)$$

An unbiased estimator of the mean response is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}. \quad (58)$$

The variance of this estimator is given by

$$\text{Var}[\hat{\mu}(\mathbf{x}_0)] = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0. \quad (59)$$

Leverage notation. Define the **hat matrix** as

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top. \quad (60)$$

Its diagonal elements h_{ii} are called **leverages** and quantify the influence of each design point on its fitted value. More generally, the leverage associated with an arbitrary design point $\mathbf{x}_0$ is defined as

$$h(\mathbf{x}_0) = \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0. \quad (61)$$

With this notation,

$$\text{Var}[\hat{\mu}(\mathbf{x}_0)] = \sigma^2 h(\mathbf{x}_0). \quad (62)$$

Replacing σ^2 with $\hat{\sigma}^2$, a $(1 - \alpha)100\%$ confidence interval for the mean response at $\mathbf{x}_0$ can be written as

$$\hat{\mu}(\mathbf{x}_0) \pm t_{\alpha/2, n-p} \hat{\sigma} \sqrt{h(\mathbf{x}_0)}. \quad (63)$$

This interval reflects the uncertainty associated with estimating the expected response and becomes narrower as the amount of data increases or as $\mathbf{x}_0$ lies closer to the center of the experimental design (typically corresponding to smaller leverage values).

6.3 Confidence Intervals on a New Response

In many practical applications, the objective is to predict a new future observation y_0 at the regressor setting $\mathbf{x}_0$. Unlike the mean response, a new observation includes both the uncertainty in the estimation of $\boldsymbol{\beta}$ and the intrinsic random error of the process.

The predicted value is

$$\hat{y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}, \quad (64)$$

and the variance of the prediction error is

$$\text{Var}[y_0 - \hat{y}_0] = \sigma^2 (1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0). \quad (65)$$

Using the leverage notation,

$$\text{Var}[y_0 - \hat{y}_0] = \sigma^2 (1 + h(\mathbf{x}_0)). \quad (66)$$

A $(1 - \alpha)100\%$ prediction interval for a new response at $\mathbf{x}_0$ is therefore given by

$$\hat{y}_0 \pm t_{\alpha/2, n-p-1} \hat{\sigma} \sqrt{1 + h(\mathbf{x}_0)}. \quad (67)$$

Prediction intervals are always wider than confidence intervals on the mean response, reflecting the additional variability due to the random error term. They are particularly relevant in design of experiments when assessing the expected variability of future experimental outcomes.

6.4 Leverage Points

The disposition of design points in the regressor space plays a fundamental role in determining the statistical properties of a regression model. In particular, observations located far from the center of the experimental region may exert a disproportionate influence on parameter estimates, fitted values, and associated summary statistics. Such observations are referred to as **high-leverage points**. The concept of leverage is conveniently described through the **hat matrix**

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad (68)$$

which maps the vector of observed responses to the vector of fitted values, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$. Under the usual assumptions of the MLR model, the variance-covariance matrices of the fitted values and residuals are given by

$$\text{Var}(\hat{\mathbf{y}}) = \sigma^2 \mathbf{H}, \quad \text{Var}(\mathbf{e}) = \sigma^2 (\mathbf{I} - \mathbf{H}), \quad (69)$$

where $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.

The diagonal elements h_{ii} of $\mathbf{H}$ quantify the leverage associated with the i -th observation and may be interpreted as the influence of y_i on its fitted value $\hat{y}_i$. Since

$$\sum_{i=1}^n h_{ii} = \text{rank}(\mathbf{H}) = \text{rank}(\mathbf{X}) = p, \quad (70)$$

the average leverage is p/n , where p is the number of estimated model parameters, including the intercept.

As a practical guideline, an observation is often considered to have high leverage if

$$h_{ii} > \frac{2p}{n}. \quad (71)$$

High-leverage points are not necessarily problematic; however, because of their potential to strongly affect model estimates and predictions, they warrant careful examination, especially in the presence of large residuals.